

## 4.2) Subspaces.

A subset  $(W)$  of a vector space  $V$  is called a subspace of  $V$ , if  $W$  itself is a vector space under addition & scalar multiplication defined on  $V$ .

**Axioms** 1- closure of  $W$  under addition.

4- existence of zero vector in  $W$ .

5- negative of every vector of  $W$  also in  $W$ .  $\therefore -a, a \in W$

6- closure of scalar multiplication.

— only need to prove 1, 6 & 4, 5 will hold as consequence.

let  $u, v$  are vectors in  $W$  which is a subset of  $V$ .

$W$  is a subspace if,

i)  $u, v \in W$  &  $u+v \in W$

ii)  $ku \in W$ ,  $u \in W, k \in \mathbb{R}$

The zero subspace:

$$\text{let } V = C(-\infty, \infty)$$

$$W = \{0\}$$

$$\Rightarrow \text{Axiom } 1 \Rightarrow f(0) = 0 = u$$

$$g(0) = 0 = v$$

$$\& \quad u + v = 0 \in W$$

$$\Rightarrow \text{Axiom } 6 \Rightarrow \text{let } k \in \mathbb{R}$$

$$k = \{1, 2, \dots\}$$

$$k \cdot u = 1(0) = 0 \in W$$

$$= 2(0) = 0 \in W$$

So,

$W$  is a subspace of  $V$  since it holds the closure law for addition & scalar multiplication.



Choose a vector which is in the subspace.

- Example 1 was the zero subspace.

Example 2: Even functions.

let  $v(-\infty, \infty)$ ,  $M = \{f(x) | f(-\infty) = f(x)\}$

$$f_1 = x^2$$

$$f_2 = \cos x$$

$$f_3 = x^4 + 3x^2 + 5$$

$$f_4 = |x|$$

Verification:

Zero vector:

$$f_1(0) = (0)^2 = 0$$

$$f_1(-x) = 0, \quad f_1(x) = 0$$

Closure under addition:

$$f_1 + f_2(x) = x^2 + \cos x$$

$$f_1 + f_2(-x) = (-x)^2 + \cos(-x)$$

$$= x^2 + \cos x = (f_1 + f_2)(x)$$

Closure under scalar multiplication:

$$f_2(x) = \cos x$$

$$K f_2(x) = 3 \cos x \quad \therefore K=3$$

$$K f_2(-x) = 3 \cos(-x) \\ = 3 \cos(x) = K f_2(x) \in W$$

So, even functions form a subspace.

2 Non subspace examples.  $V = xy\text{-plane}$

vectors with +ve  $x$ .

$$\text{subset } W = \{ (x, y, z) \in \mathbb{R}^3 \mid x > 0 \}$$

Why its not a subspace:

$$1) \text{ zero vector } \Rightarrow W(0, 0, 0) \notin W$$

$$\text{Since } x \neq 0$$

2) scalar multiplication:

$$\text{if scalar } < 0, \text{ let } K = -1$$

$$W(x, y, z) \cdot (-1) \notin W$$

So, does not hold closure for multiplication with scalar.



### 3) subspace matrices in $\mathbb{R}^3$

$V$  is a space for all  $3 \times 3$  matrices.

let  $W$  be a subset that contains all lower triangular matrices.

$$\text{let } v_1 = \begin{Bmatrix} 1 & 0 & 0 \\ 3 & 5 & 0 \\ 0 & 1 & 7 \end{Bmatrix} \quad v_2 = \begin{Bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 10 & 9 \end{Bmatrix}$$

Zero matrix  $\Rightarrow W[0,0] \in W$

1) closure under addition:

$$v_1 + v_2 = \begin{Bmatrix} 2 & 0 & 0 \\ 3 & 7 & 0 \\ 0 & 11 & 16 \end{Bmatrix} \in W$$

$v_1 + v_2$  is a lower triangular matrix so  $\in W$ .

2) scalar multiplication: let  $k=2$

$$2v_1 = \begin{Bmatrix} 2 & 0 & 0 \\ 6 & 10 & 0 \\ 0 & 2 & 14 \end{Bmatrix} \text{ which } \in W$$

so,  $W$  is a

subspace.

4) Example of ordered pairs  
in  $\mathbb{R}^3$

let  $V \Rightarrow \mathbb{R}^3$  plane  
~~#~~  $W$  is a subspace within  
 $xy$ -plane.

$$v_1 = (1, 2, 0)$$

$$v_2 = (1, -1, 0)$$

$$v_3 = (2, 3, 0)$$

$$\text{zero vector} \Rightarrow W (0, 0, 0) \\ \in W (0, 0, 0)$$

closure in addition:

$$v_2 + v_3 = (3, 2, 0)$$

which is still in  $W$ .

scalar multiplication:

$$4v_1 = (4, 8, 0) \in W$$

So,  $xy$ -plane is  $\mathbb{R}^3$   
 is a subspace.